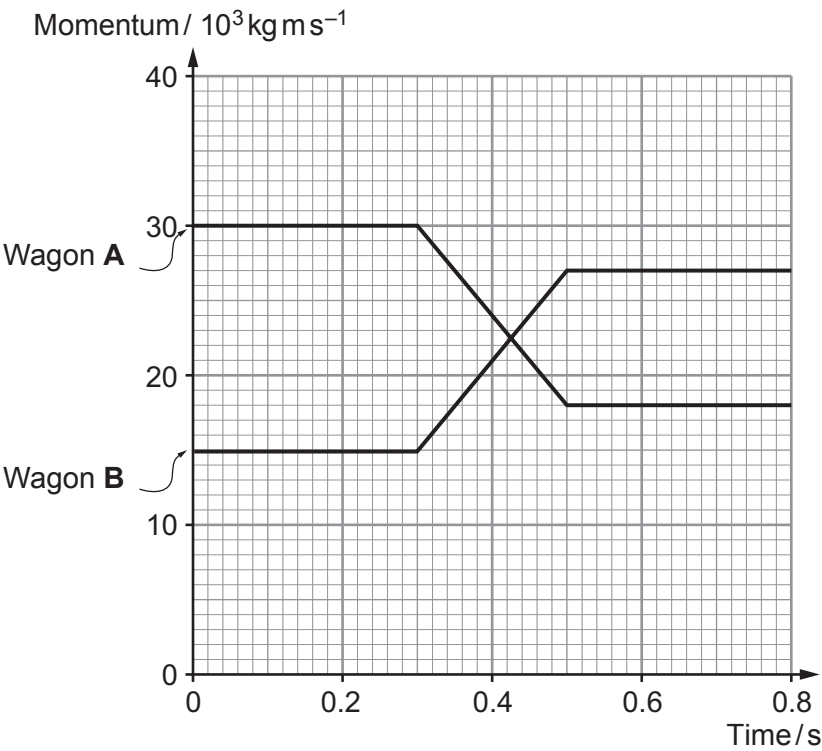


7. (a) Momentum is a *vector*. State what is meant by a vector. [1]

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(b) The graph shows how the momentum of two colliding railway wagons (**A** and **B**) varies with time. The collision takes place between 0.30 s and 0.50 s as shown. The wagons remain joined together after impact.



(i) Explain with suitable calculations how the graphs show that external forces on the system are negligible. [3]

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(ii) Calculate the final velocity of the two wagons given that the total mass of wagon **A** and wagon **B** is 25 000 kg. [3]

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(c) (i) State Newton’s Second Law of motion in terms of momentum. [2]

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(ii) Determine from the graph the resultant force on wagon **A** during the collision. [3]

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(iii) The force experienced by wagon **B** during the collision is equal and opposite to the force experienced by wagon **A**. State which law of motion this is an example of **and** explain how the graph confirms this law. [2]

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END OF PAPER

